Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior
using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.
You are tasked with analyzing the company's consumer behavior dataset to answer the following overarching business question:
"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

3. Dataset Summary

customer_shopping_behavior

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

4. Exploratory Data Analysis using Python

- Data loading using pandas

```
[28]: #pip install pandas
[29]: import pandas as pd
[30]: df = pd.read_csv("customer_shopping_behavior.csv")
```

- Initial exploration using info() and describe()

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

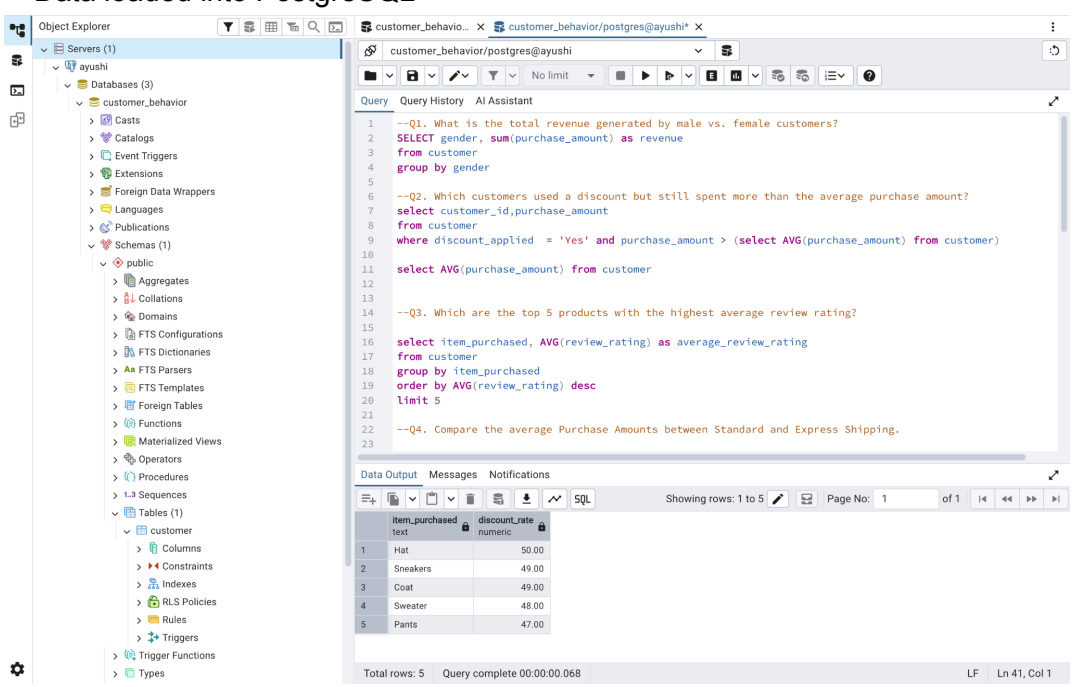
- Column Standardization: Renamed columns to snake case for better readability and documentation.

- Feature Engineering:

  - Created age_group column by binning customer ages.
  - Created purchase_frequency_days column from purchase data.

- Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

- Data loaded into PostgreSQL



5. Data Analysis using SQL

- Revenue by gender

```
SELECT gender, sum(purchase_amount) as revenue
from customer
group by gender
```

	gender text	revenue numeric
1	Female	75191
2	Male	157890

- High-spending discount users

```
select customer_id,purchase_amount
from customer
where discount_applied  = 'Yes' and purchase_amount > (select AVG(purchase_amount) from customer)
```

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79

- Top 5 products by rating

```
select item_purchased, AVG(review_rating) as average_review_rating
from customer
group by item_purchased
order by AVG(review_rating) desc
limit 5
```

	item_purchased text	average_review_rating double precision
1	Gloves	3.8614285714285725
2	Sandals	3.8443750000000003
3	Boots	3.8187500000000005
4	Hat	3.8012987012987005
5	Skirt	3.784810126582278

- Shipping type comparison

```
select shipping_type, avg(purchase_amount) as average_purchase_amount
from customer
where shipping_type in ('Standard','Express')
group by shipping_type
```

	shipping_type text	average_purchse_amount numeric
1	Standard	58.4602446483180428
2	Express	60.4752321981424149

- Subscribers vs non-subscribers

```
select subscription_status,count(customer_id),sum(purchase_amount) as
Revenue,AVG(purchase_amount) as avg_spend
from customer
group by subscription_status
order by avg_spend desc
```

	subscription_status text	count bigint	revenue numeric	avg_spend numeric
1	No	2847	170436	59.8651211801896733
2	Yes	1053	62645	59.4919278252611586

- Discount-dependent products

```
select item_purchased,
ROUND(100*SUM(CASE WHEN discount_applied ='Yes' THEN 1 ELSE 0 end)/count(*),2) as
discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5
```

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

- Customer segmentation

```
with customer_type as
(
  select customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 and 10 THEN 'Returning'
      ELSE 'Loyal'
    END as customer_segment
  from customer
)

select customer_segment,count(*)
from customer_type
group by customer_segment
```

	customer_segment text	count bigint
1	Loyal	3116
2	New	83
3	Returning	701

- Top products per category

```
with item_counts as (
select category,
item_purchased,
count(customer_id) as Total_orders,
ROW_NUMBER() over (partition by category order by count(customer_id) DESC) as item_rank
from customer
group by category,
item_purchased)
select * from item_counts where item_rank<4
```

	category text	item_purchased text	total_orders bigint	item_rank bigint
1	Accessori...	Jewelry	171	1
2	Accessori...	Sunglasses	161	2
3	Accessori...	Belt	161	3
4	Clothing	Blouse	171	1
5	Clothing	Pants	171	2
6	Clothing	Shirt	169	3
7	Footwear	Sandals	160	1
8	Footwear	Shoes	150	2
9	Footwear	Sneakers	145	3
10	Outerwear	Jacket	163	1
11	Outerwear	Coat	161	2

- Repeat buyers and subscriptions

```
select subscription_status,count(customer_id) as Total_customers from customer
where previous_purchases >5
GROUP BY subscription_status
```

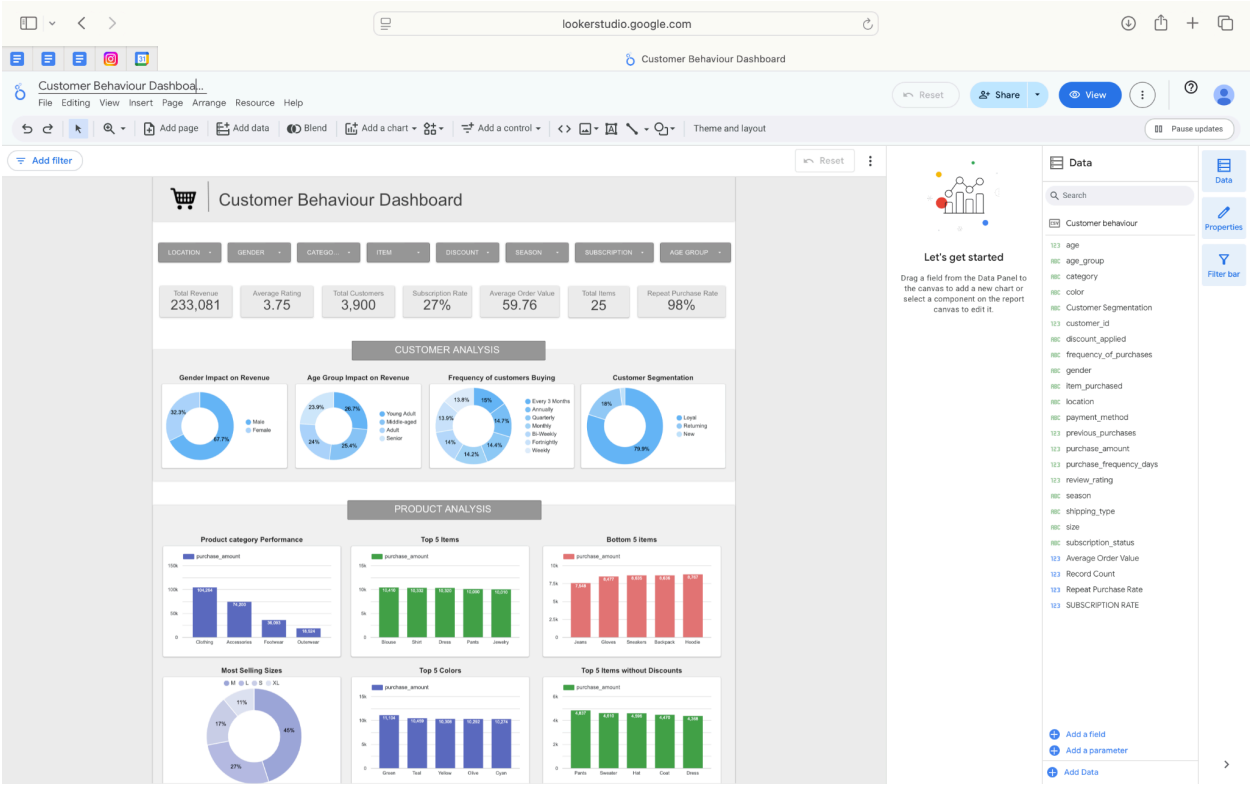
	subscription_status text	total_customers bigint
1	No	2518
2	Yes	958

- Revenue by age group

```
select  age_group,sum(purchase_amount) as Total_revenue
from customer
group by age_group
order by Total_revenue desc
```

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

6. Dashboard in Lookerstudio

Created an interactive dashboard to visualize insights

Link to the Dashboard : https://lookerstudio.google.com/s/uLooX4eMXWo